WASP-43b

R-0349

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-43_260131` · created 2026-07-18T09:29:31+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461071.8856 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.167 +/- 0.026
Transit depth (Rp/Rs)^2	2.78 +/- 0.87 [%]
Orbital Inclination (inc)	80.25 +/- 2.34
Airmass coefficient 1 (a1)	0.861 +/- 0.012
Airmass coefficient 2 (a2)	0.108 +/- 0.059
Scatter in the residuals of the lightcurve fit is	1.36 %
Best Comparison Star	#3 - [175, 45]
Optimal Method	PSF photometry
Transit Duration (day)	0.042 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.167 ± 0.026 vs 0.1594 (deviation 0.29σ)
Duration fit vs published	0.042 d vs 0.0512 d (deviation 0.71σ)
Mid-time O–C	6.7 min
Depth SNR	6.4
Residual scatter	1.36 %

Light curve

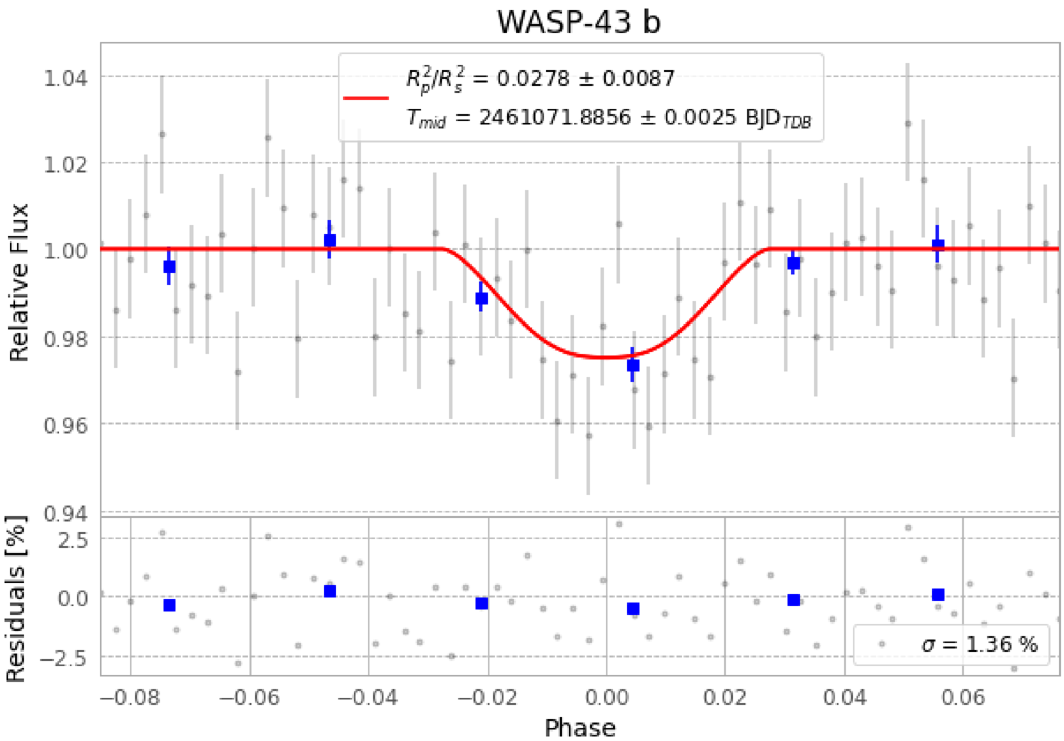

Figure 1 — FinalLightCurve_WASP-43 b_2026-01-31.png (SHA-256 dc31cb5880630533…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [471, 170], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 6c7923708bb92082…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8b944da`

Data provenance

64 FITS frames (42 MB), `WASP-43260131072715.FITS` → `WASP-43260131103616.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-260131.log (b609ca59d0a5…), FinalLightCurve_WASP-43 b_2026-01-31.png (dc31cb588063…), FinalLightCurve_WASP-43 b_2026-01-31.csv (0c3a3f4cc634…)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 09:29Z.